

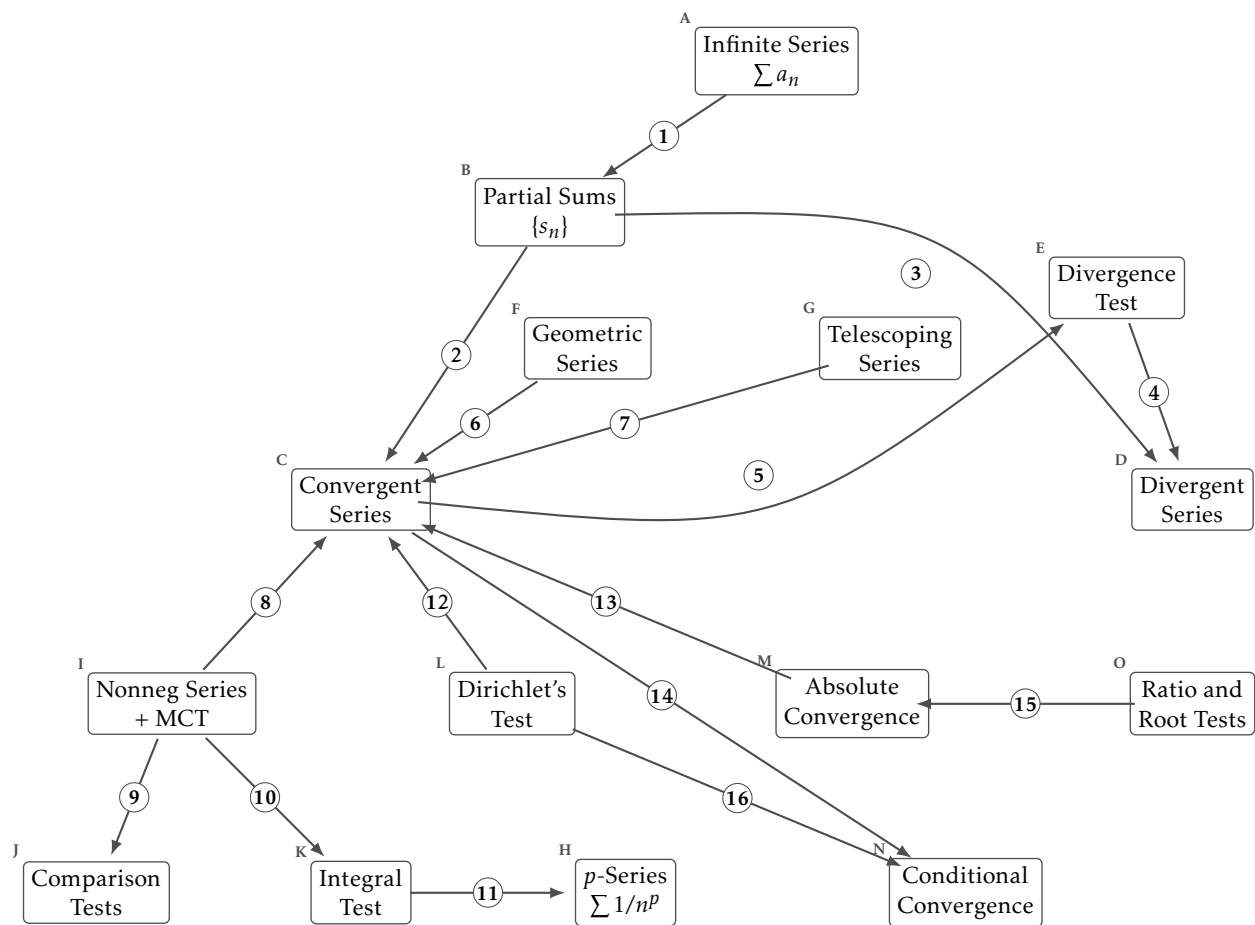
Infinite series: answers

Math 163 • Prof. Chowdhury

Every arrow, with the reason underneath. Keep this away from the blank worksheet.

What each box means

- A. **Infinite Series** $\sum a_n$. The expression $a_1 + a_2 + a_3 + \cdots$. It has no meaning until the partial sums give it one.
- B. **Partial Sums** $\{s_n\}$. $s_n = a_1 + \cdots + a_n$. An ordinary sequence, so everything you know about sequences applies to it.
- C. **Convergent Series**. $\sum a_n$ converges when $\{s_n\}$ has a finite limit, and the sum is that limit.
- D. **Divergent Series**. $\sum a_n$ diverges when $\{s_n\}$ fails to converge, whether it runs away or oscillates.
- E. **Divergence Test**. If $a_n \not\rightarrow 0$ then $\sum a_n$ diverges. It is a test for divergence only, and it never certifies convergence.
- F. **Geometric Series**. $\sum ar^n$, which converges exactly when $|r| < 1$, with sum $a/(1 - r)$ and partial sums $a(1 - r^n)/(1 - r)$.
- G. **Telescoping Series**. $\sum (b_n - b_{n+1})$, whose partial sums collapse to $b_1 - b_{n+1}$.
- H. **p-Series** $\sum 1/n^p$. Converges exactly when $p > 1$. The boundary $p = 1$ is the harmonic series, which diverges.
- I. **Nonneg Series + MCT**. If $a_n \geq 0$ the partial sums increase, so the monotone convergence theorem turns convergence into boundedness.
- J. **Comparison Tests**. Direct: $0 \leq a_n \leq cb_n$ eventually and $\sum b_n$ convergent gives $\sum a_n$ convergent. Limit: $\lim a_n/b_n \in (0, \infty)$ gives the same behaviour for both.
- K. **Integral Test**. If f is positive, continuous and eventually decreasing with $f(n) = a_n$, then $\sum a_n$ and $\int_1^\infty f$ converge or diverge together.
- L. **Dirichlet's Test**. If a_n decreases to 0 and the partial sums of b_n are bounded, then $\sum a_n b_n$ converges. Proved by summation by parts.
- M. **Absolute Convergence**. $\sum a_n$ converges absolutely when $\sum |a_n|$ converges. This is strictly stronger than convergence.
- N. **Conditional Convergence**. $\sum a_n$ converges conditionally when it converges and $\sum |a_n|$ does not. The convergence rests on cancellation.
- O. **Ratio and Root Tests**. With $L = \lim |a_{n+1}/a_n|$ or $L = \limsup |a_n|^{1/n}$: absolutely convergent if $L < 1$, divergent if $L > 1$, and no conclusion if $L = 1$.



The arrows

1. Convergence of the series is defined to be convergence of its sequence of partial sums. Nothing else defines it. Every test in this chapter is a statement about the partial sums, reached without ever computing them.
2. If the partial sums have a finite limit then the series converges, and its sum is that limit. The definition read from left to right.
3. If the sequence of partial sums has no finite limit then the series diverges, whether the partial sums are unbounded or oscillate without approaching a limit. Divergence is the negation of convergence, so an unbounded sequence of partial sums and a bounded oscillating one are both divergent.
4. If $a_n \not\rightarrow 0$ then the partial sums cannot converge, so the series diverges. The difference $s_n - s_{n-1}$ is a_n . If s_n had a limit, that difference would have to tend to zero.
5. A convergent series has terms tending to zero, and the divergence test is that theorem read in contrapositive form. It is one statement with two readings. $s_n - s_{n-1} = a_n$, so if s_n converges then $a_n \rightarrow 0$. The condition is necessary and not sufficient: $1/n \rightarrow 0$ and $\sum 1/n$ diverges, so the contrapositive detects divergence and never certifies convergence.
6. A geometric series converges exactly when its ratio has modulus below one, and then the closed form for the partial sums gives the sum. One of the two series here with partial sums in closed form, which is what makes it a benchmark for the comparison tests.
7. In a telescoping series consecutive terms cancel, so the partial sums collapse and converge exactly when b_n does. The other series whose partial sums you can write down. The sum is b_1 minus the limit of b_n .
8. For a series of nonnegative terms the partial sums increase, so the series converges exactly when they are bounded above. Every test for a nonnegative series rests on this. It replaces the question of a limit with the question of an upper bound, and an upper bound can be established without knowing the sum.
9. Bounding a_n term by term against a series of known behaviour bounds the partial sums of $\sum a_n$, which is what the comparison tests do. Direct comparison bounds the partial sums outright. Limit comparison instead asks for $\lim a_n/b_n$ to lie in $(0, \infty)$, which is enough to give both series the same behaviour.
10. For a positive decreasing f with $f(n) = a_n$, the rectangles of the partial sums are bounded above and below by $\int f$, so the series and the improper integral converge together. The rectangles drawn on either side of the graph give the two inequalities, and both of them need f positive, continuous and eventually decreasing.
11. Applying the integral test to $1/x^p$ determines exactly which exponents give convergence: $\sum 1/n^p$ converges if and only if $p > 1$. $\int_1^\infty x^{-p} dx$ is finite exactly when $p > 1$, and at $p = 1$ the series is harmonic and diverges. It is the only test on this map that decides the question: the ratio and root tests both give $L = 1$ for every p , and a comparison would need a benchmark, which for a p -series means another p -series.
12. If a_n decreases to zero and the partial sums of b_n are bounded then $\sum a_n b_n$ converges, and the alternating series test is the case $b_n = (-1)^n$. The proof is summation by parts, which is the discrete version of integration by parts. For a case that is not alternating at all, look at $\sum \sin(n\theta)/n$.
13. If $\sum |a_n|$ converges then $\sum a_n$ converges, so absolute convergence is the stronger property. The Cauchy criterion gives it: by the triangle inequality the tails of $\sum |a_n|$ control the tails of $\sum a_n$.
14. A series for which $\sum a_n$ converges and $\sum |a_n|$ diverges is what conditional convergence means. The alternating harmonic series is the case to hold on to: $\sum (-1)^{n+1}/n$ converges to $\ln 2$ while $\sum 1/n$ diverges. The convergence depends on cancellation between the positive and negative terms, and taking absolute values removes it.
15. The ratio and root tests compare $|a_n|$ against a geometric series, so what they establish is convergence of $\sum |a_n|$. Both are computed from $|a_n|$, and both are inconclusive when the limit is 1: $\sum 1/n$ diverges and $\sum 1/n^2$ converges, and both give 1.
16. Dirichlet's test establishes convergence from the sizes and signs of the terms together, so it can certify a series for which $\sum |a_n|$ diverges. The alternating harmonic series again: Dirichlet's test gives the convergence of $\sum (-1)^{n+1}/n$, and $\sum 1/n$ diverges.

Claims that look right and are not

- A series of nonnegative terms converges exactly when its terms tend to zero.
- Bounded partial sums are enough for a series to converge.
- Rearranging the terms of a convergent series never changes its sum.
- The integral test applies to any series whose terms are positive.
- A series that passes the ratio test at $L = 1$ diverges.

Counterexamples and benchmarks

- The harmonic series $\sum 1/n$ diverges even though its terms tend to zero.
- The alternating harmonic series $\sum (-1)^{n+1}/n$ converges conditionally, to $\ln 2$.
- $\sum \sin(n\theta)/n$ converges by Dirichlet's test and is not alternating.
- Summation by parts is the discrete integration by parts, and it is what proves Dirichlet's test.
- The Cauchy criterion for series: control every far enough tail at once, not one chosen tail.

One last question

Which single arrow carries the most of this chapter? There is more than one defensible answer, and the argument is the useful part.